

4 (a) A car headlamp is marked 12 V, 36 W. It is switched on for a 20 minute journey.

Calculate

(i) the current in the lamp,

current = A [2]

(ii) the charge which passes through the lamp during the journey,

charge = C [2]

(iii) the energy dissipated by the lamp during the journey ,

energy dissipated = J [1]

(iv) the resistance of the lamp.

resistance = Ω [1]

- (b) Three of the headlamps in (a) are connected to a battery of e.m.f. 12 V and negligible internal resistance as shown in Fig. 4.1.

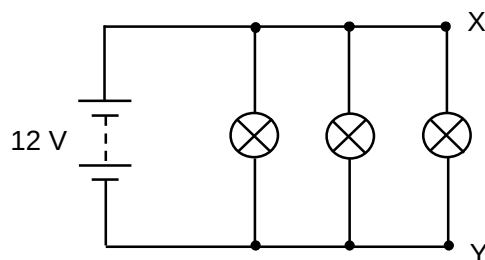


Fig. 4.1

- (i) Determine the current provided by the battery.

current = A [2]

- (ii) Another identical headlamp is connected across points X and Y in the circuit.

1. Describe and explain any change to the brightness of the three headlamps in Fig. 4.1.

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 [2]

2. The battery is replaced by another of the same e.m.f. but with an internal resistance of $1.0 \, \Omega$.

State and explain any change to your answer in **part 1**.

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 [2]

5 Fig. 5.1 shows a 1.6 m long solenoid with 400 turns and a cross-sectional diameter of